

Longifolene enhances *Beauveria bassiana* bait by counteracting repellency in the fungus-growing termite *Odontotermes formosanus*

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Abstract

BACKGROUND: The fungus-growing termite *Odontotermes formosanus* is a significant pest of hydraulic infrastructure, driving the need for targeted and environmentally sound control strategies. The entomopathogenic fungus *Beauveria bassiana* is a promising biocontrol candidate against *O. formosanus*, but its efficacy in bait formulations can be reduced by termite repellence behavior induced by fungal volatiles. This study explored the potential of the plant-derived volatile longifolene to counteract this repellence or avoidance and enhance bait performance.

RESULTS: In laboratory olfactometer assays, dry conidia of *B. bassiana* repelled termite workers in a dose-dependent manner, whereas longifolene at 0.2–0.4 mmol mL⁻¹ acted as a significant attractant. In toxic bait tests, incorporating longifolene reversed the repellency of a low conidial concentration of 1×10^7 conidia mL⁻¹, resulting in an attractant rate > 70%. This combined bait was also proved effective in field trials at reservoir sites, attracting significantly more workers than control baits.

CONCLUSION: Our findings indicate that longifolene can mitigate the olfactory repellence or avoidance of *B. bassiana* conidia in baits by the termites, facilitating an 'attract-and-infect' approach. This strategy offers a novel, sustainable option for managing *O. formosanus*.

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Supporting information may be found in the online version of this article.

Keywords: *Beauveria bassiana*; fungus-growing termite; longifolene; repellence; toxic bait

1 INTRODUCTION

Odontotermes formosanus (Shiraki) (Blattodea: Termitidae), commonly referred to as the black-winged subterranean termite, is a notorious pest that causes substantial economic damage to hydraulic infrastructure globally.^{1,2} As a typical subterranean termite and a fungus-growing termite, it excavates extensive tunnels in reservoir dams, dikes, and irrigation systems, resulting in structural instability, leakage, and even catastrophic failures.³ The termite's foraging and avoidance behaviors are cryptic and widespread.⁴ This limits the efficiency of conventional control methods. Therefore, targeted and environmentally sustainable management strategies are needed to effectively tackle termites. *Odontotermes formosanus* is a highly destructive fungus-growing termite with strong foraging selectivity and chemical sensitivity,⁵ making it an important model for bait-based management studies. Its reliance on olfactory cues during foraging suggests that bait effectiveness may be strongly affected by volatile-mediated attraction or avoidance. Given that worker behavior determines bait discovery, acceptance, and pathogen spread within the colony, understanding worker responses is essential for developing effective bait-based management strategies. Therefore, targeting workers is crucial not only for maximizing bait uptake but also for enabling colony-level dissemination of

the entomopathogen, making worker-focused interventions particularly important for sustainable control of *O. formosanus*.

Entomopathogenic fungi have emerged as promising biological control agents for termite management. They are valued for their host specificity and ecological safety. Among these, *Beauveria bassiana* (Balsamo) Vuillemin has been extensively investigated for its pathogenicity against various termite species, including

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O. formosanus.^{6,7} Laboratory and field trials have confirmed that *B. bassiana* conidia can infect and kill termite workers, with mortality resulting from fungal hyphal penetration and toxin production.^{8,9} However, the practical efficacy of *B. bassiana* is often limited by its inherent biological characteristics.^{6,10}

Beauveria bassiana has attracted increasing attention as a bait-compatible biocontrol agent,^{8,11} because termite management often depends on delivering toxicants through foraging behavior rather than direct contact sprays. In bait systems, infected individuals can potentially transfer conidia to nestmates through social interactions, making fungal baits particularly suitable for cryptic subterranean termites.¹² However, practical deployment of *B. bassiana* in termite baits remains challenging because termites may detect fungal volatiles and avoid treated food before ingestion. This problem is especially important for fungus-growing termites, whose foraging decisions are strongly shaped by chemical cues and feeding preferences. Therefore, improving bait attractiveness while preserving fungal infectivity is a key step toward developing effective *B. bassiana*-based termite management strategies. Against this background, identifying a volatile cue that can attract termites and offset fungal avoidance may improve the practical performance of bait-based biocontrol.

A critical challenge in the fungal biocontrol of termites is the avoidance or repellency of volatile compounds emitted by entomopathogenic fungi.^{6,8,13} Studies have demonstrated that *B. bassiana* isolates emit specific volatile blends (e.g., phenol, 3-octanol, and 2-nonanol) that termites can detect via olfactory cues, triggering avoidance behavior.^{8,14} This repellency therefore reduces termite contact with fungal inocula, thereby impairing control efficacy. Notably, *B. bassiana*'s repellent activity is concentration-dependent;⁶ however, the threshold concentration at which repellency becomes negligible, that is, a concentration that preserves pathogenicity without deterring termite foraging remains unclear, and consequently impeding the optimization of fungal formulations.

In nature, some plant species are highly attractive to termites, and their volatile components have been explored as potential lures to enhance baiting systems.^{15,16} For instance, longifolene – a tricyclic sesquiterpene abundant in the volatile emissions of *Pinus densiflora* and *Pinus thunbergii*,¹⁷ has garnered attention due to its close association with termite-preferred host plants. Previous studies have reported anti-termite activity of longifolene autooxidation products. However, the potential attractant effect of longifolene itself on *O. formosanus* has not been systematically explored. In particular, the optimal concentration for inducing foraging behavior remains unclear.

Additionally, the interactive effect between longifolene and *B. bassiana* represents an unaddressed research gap. We hypothesized that, if longifolene acts as a termite attractant, combining it with *B. bassiana* in toxic baits could counteract fungal repellency, and consequently increase termite ingestion of conidia. However, it remains unknown whether this combination would synergistically improve control efficacy or if longifolene might interfere with fungal viability or pathogenicity (e.g., by inhibiting conidial germination or hyphal growth). To address these gaps, this study aimed, (i) to quantify the dose-dependent repellent response of *O. formosanus* workers to *B. bassiana* conidia and identify a concentration with minimal repellency; (ii) to evaluate the attractant activity of longifolene and determine its optimal concentration for luring termites; and (iii) to assess the combined use of longifolene and *B. bassiana* in toxic baits, verifying whether the attractant could mitigate fungal repellency without compromising its

viability or pathogenicity. The findings of this research initiative aim to inform the development of novel, efficient bait formulations for sustainable termite management.

2 METHODS AND MATERIALS

2.1 Inhibitory effect of longifolene on the growth and virulence-related characteristics of *B. bassiana*

A gradient solution of longifolene ($\geq 98\%$ purity; Aladdin, Pudong, China) was prepared in triethyl citrate (TEC) and subsequently diluted to yield a series of working solutions at final concentrations of 0.02, 0.04, 0.08, 0.16, 0.32 and 0.64 mmol mL⁻¹. For each concentration, 0.2 mL of the longifolene working solution was thoroughly mixed into 49.8 g of molten potato dextrose agar (PDA), resulting in a final concentration of longifolene in the medium. A control medium was prepared by adding an equal volume of dimethyl sulfoxide (DMSO, without longifolene) to the base. After homogenization, each treatment and the control medium were dispensed into ten sterile 9-cm Petri dishes.

2.1.1 Assay of radial growth

A freshly harvested 0.5 cm mycelial plug from a 7-day-old actively growing *B. bassiana* BJ03 colony (isolated from *Apriona germari*) was aseptically placed at the center of each dish. All plates were then sealed, inverted, and incubated at $25 \pm 2^\circ\text{C}$ under complete darkness. After 10 days of incubation, the colony diameter of *B. bassiana* was measured in two perpendicular directions using the cross method. The mean colony diameter for each treatment was calculated and compared.

2.1.2 Test of conidial germination

Conidial germination rate was determined using the hanging drop method. Briefly, 10 μL of conidial suspension (1×10^6 conidia mL⁻¹) prepared from 10-day-old colonies grown on longifolene-amended or control PDA was placed on a sterile glass coverslip. The coverslip was inverted over a cavity slide containing a drop of sterile water to maintain humidity, and incubated at $25 \pm 2^\circ\text{C}$ for 24 h. Germination was defined as the emergence of a germ tube longer than the conidial diameter. At least 300 conidia were counted per replicate, with five replicates per treatment.

2.1.3 Test of virulence against termite workers

Healthy, active *O. formosanus* workers (second to third instar) were selected for virulence assays following a 24 h acclimatization period. Conidia were harvested from 10-day-old *B. bassiana* colonies grown on longifolene-amended (0.02–0.64 mmol mL⁻¹) or control PDA, and adjusted to a concentration of 1×10^8 conidia mL⁻¹ using sterile distilled water containing 0.05% (v/v) Tween-80. A blank control (sterile water + 0.05% Tween-80) was also included.

The bioassay was conducted using a filter paper contact method. Briefly, a 9 cm sterile filter paper was placed in a Petri dish and moistened with 5 mL of the corresponding conidial suspension or blank control. Twenty workers were introduced into each dish and incubated at $25 \pm 2^\circ\text{C}$, $90\% \pm 5\%$ relative humidity (RH) under complete darkness. Mortality was recorded at the seventh day. Dead termites were surface-sterilized with 75% ethanol for 30 s, rinsed three times with sterile water, and placed on moist filter paper at $25 \pm 2^\circ\text{C}$ to confirm fungal infection by observing hyphal growth on the cadaver surface. Each treatment

was replicated ten times. Corrected mortality was calculated using Abbott's formula:

$$\text{Corrected mortality (\%)} = \frac{\text{Confirmed fungal infected mortality}}{\text{Total number} - \text{Blank control mortality}} \times 100$$

2.2 Repellency of *B. bassiana* conidia on *O. formosanus* workers

Workers of *O. formosanus* were collected from field nests using a modified baiting method.¹⁸ Briefly, traps containing sugarcane bagasse were placed overnight, after which captured termites were transported to the laboratory. The colony was maintained under controlled conditions ($25 \pm 2^\circ\text{C}$, $90\% \pm 5\%$ RH, complete darkness) in Petri dishes lined with moist filter paper and provided with sterilized sugarcane bagasse and pine wood as food. Healthy, active workers of similar size were selected for bioassays following a 24 h acclimatization period. The *B. bassiana* strain was cultured on PDA at $25 \pm 2^\circ\text{C}$ for 14 days. Conidia were harvested, suspended in sterile distilled water containing 0.05% (v/v) Triton X-100, and enumerated using a hemocytometer. The suspension was serially diluted to obtain final concentrations of 1×10^6 , 1×10^7 , 1×10^8 , 1×10^9 , and 1×10^{10} conidia mL^{-1} . Conidial viability was assessed prior to experiments, and their germination exceeded 90%. For bait preparation, sterilized sugarcane bagasse (2–3 mm particle size) was impregnated with 30.0 mL of the conidial suspension of each concentration (or sterile water for controls) and mixed thoroughly. The baits were air-dried in a laminar flow hood to a moist and non-dripping state. Repellency was assessed using a Y-tube olfactometer.⁶ One arm (treatment arm) contained a sample of the fungal bait (2.0 g, wrapped in 40-mesh nylon gauze), while the other (control arm) contained an equal amount of control bait. Airflow was maintained at 300 mL min^{-1} . A visual 'push-pull' design was employed: the release zone was illuminated, while the choice arms were shaded to stimulate photonegative movement. For each trial, 20 workers were introduced into the release zone, and their distribution was recorded at 10-min intervals over 90 min. The repellency rate was calculated as:¹⁹

$$\text{Repellency rate (\%)} = \frac{P_c - P_t}{P_c + P_t} \times 100$$

where P_c represents the average proportion of termites in the control zone; P_t the average proportion of termites in the treatment zone.

Each replicate lasted for 90 min of observation period, and each termite worker was used only once and then discarded to avoid pseudoreplication.

2.3 Attractant activity of longifolene on *O. formosanus* workers

The attractant effect of longifolene on termite workers was evaluated using a closed-system Y-tube olfactometer. Longifolene was dissolved in TEC to create five test concentrations (0.02, 0.04, 0.08, 0.16, and $0.32 \text{ mmol mL}^{-1}$). Sterilized sugarcane bagasse was used as a carrier, soaked in the test solutions or pure TEC (control), and air-dried to a moist state. In the olfactometer, one arm (treatment arm) received volatiles from longifolene-treated bagasse, while the other (control arm) received volatiles from the TEC

control, with purified air flowing at 300 mL min^{-1} . A 'push-pull' visual design (lighted release zone, darkened choice arms) was employed to guide termite movement. For each replicate, 20 workers were released and their distribution in the arms was recorded over 10 min. The attractant rate was calculated as:¹⁹

$$\text{Attractant rate (\%)} = \frac{A_t - A_c}{A_t + A_c} \times 100$$

where A_t represents the mean number of termites in the treatment arms; A_c the mean number of termites in the control arms.

Each replicate lasted for 90 min of observation period, and each termite worker was used only once and then discarded to avoid pseudoreplication.

2.4 Synergistic attraction/repellent of longifolene combined with *B. bassiana* to *O. formosanus* workers

The synergistic attraction of longifolene combined with *B. bassiana* conidia was assessed using a dual-choice Y-tube olfactometer. Ten combined treatments were tested, comprising two *B. bassiana* conidial concentrations (1×10^7 and 1×10^9 conidia mL^{-1}) and five longifolene concentrations (0.02, 0.04, 0.08, 0.16, and $0.32 \text{ mmol mL}^{-1}$). A blank control bait, consisting of 2 g of sterilized sugarcane bagasse with 1.5 mL of sterile distilled water containing 0.05% Triton X-100 and 1 mL of TEC. For each treatment, the toxic bait was prepared by sequentially adding 1.5 mL of the *B. bassiana* conidial suspension and 1 mL of the longifolene solution to 2 g of sterile bagasse. The control or treatment bait was artificially mixed for uniformity and transferred into a nylon gauze, respectively. In the olfactometer, volatiles from the combined bait were introduced into the treatment arm, while volatiles from the blank control were introduced into the control arm, with purified air flowing at 300 mL min^{-1} . A visual 'push-pull' design was employed, with an illuminated release zone and darkened choice arms. For each replicate, 20 healthy workers were released, and their distribution was recorded over 10 min. Ten replicates were performed per treatment, with thorough cleaning of the apparatus between trials. For treatments containing the lower fungal concentration (1×10^7 conidia mL^{-1}), the attractant rate was calculated to assess the ability of the bait of affecting termite behavior. Conversely, for treatments with the higher fungal concentration (1×10^9 conidia mL^{-1}), the repellency rate was calculated, as this concentration was expected to induce avoidance behavior.

Each replicate lasted for 90 min of observation period, and each termite worker was used only once and then discarded to avoid pseudoreplication.

2.5 Field evaluation

Field trials were conducted in reservoir areas with known and active *O. formosanus* infestations based on pre-survey inspections (Table S1). Prior to bait deployment, each candidate site was examined for the presence of active termite tunnels, foraging traces, mud tubes, and termite workers, and the infestation status and termite workers were recorded. Based on prior laboratory olfactometry results, two optimal combined baits were selected for field testing and validation: Toxic Bait 1 (the formulation with the highest attractant rate at 1×10^7 conidia mL^{-1} *B. bassiana*) and Toxic Bait 2 (the formulation with the highest repellent rate at 1×10^9 conidia mL^{-1} *B. bassiana*). A blank control bait (sugarcane bagasse without *B. bassiana* and longifolene) served as

the reference/standard. The field test of longifolene-supplemented toxic baits aimed to evaluate whether longifolene counteracts repellency in the field. The experiment was replicated across three independent reservoirs, Houjing Reservoir (R1, 117.611646E, 24.028184 N), Houpan Reservoir (R2, 117.843376E, 24.679090 N) and Zumalin Reservoir (R3, 117.569478E, 23.971773 N). At each reservoir, five test points were established for a dual-choice comparison between Toxic Bait 1 and the control, and another five points for Toxic Bait 2 *versus* the control. At each test point, two permeable bait boxes – one containing the target toxic bait and the other the control bait – were buried 20 cm deep, centered on an active infested termite site and spaced 50 cm apart to minimize interference. Each box contained 10 g of the corresponding bait material. After a 20-day exposure period, all boxes were retrieved, and the number of *O. formosanus* workers in each was counted. The feeding attraction percentage for each test point was calculated as:

$$\text{Feeding attraction (\%)} = \frac{N_{\text{toxic}}}{N_{\text{control}} + N_{\text{toxic}}} \times 100$$

where N_{toxic} represents the worker counts in the toxic bait boxes; N_{control} the worker counts in the control bait boxes.

2.6 Statistical analysis

All experiments were conducted with the indicated number of replicates, and data are presented as mean $\pm$ standard deviation (SD). Prior to analysis, percentage data were arcsine square-root transformed when necessary to meet the assumptions of normality and homogeneity of variance. Differences among treatments were analyzed using one-way analysis of variance (ANOVA) when a single factor was involved, and two-way ANOVA when treatment and time or choice side were considered simultaneously. When significant effects were detected, Tukey's honest significant difference (HSD) *post hoc* test was used for multiple comparisons. Statistical significance was set at $P < 0.05$. All analyses were performed using GraphPad Prism 10.1 (GraphPad Software, San Diego, CA, USA).

3 RESULTS

3.1 Inhibitory effect of longifolene on the growth of *B. bassiana*

Longifolene induced a progressive, dose-dependent inhibition of *B. bassiana* mycelial growth (Fig. 1(i)). Colony growth inhibition differed significantly among treatments ($F_{(6,63)} = 151.7$, $P < 0.0001$). At 0.02 and 0.04 mmol mL⁻¹, the inhibition rates were 2.66% $\pm$ 1.10% and 6.56% $\pm$ 6.39%, respectively, and did not differ significantly from each other ($P = 0.9171$). A marked increase to 25.59% $\pm$ 7.80% was observed at 0.08 mmol mL⁻¹, followed by further increases at 0.16 mmol mL⁻¹ (42.48% $\pm$ 9.28%), 0.32 mmol mL⁻¹ (57.62% $\pm$ 8.39%) and 0.64 mmol mL⁻¹ (77.67% $\pm$ 6.32%) ($P < 0.0001$). These results indicate that longifolene progressively suppressed fungal colony growth as concentration increased.

In contrast, longifolene had no significant effect on conidial germination (Fig. 1(ii)). One-way ANOVA confirmed the overall significance of this effect ($F_{(6,28)} = 8.287$, $P < 0.0001$). Germination rates ranging from 89.3% $\pm$ 2.44% to 92.80% $\pm$ 1.54% when the concentration of longifolene at the range of 0.02 to 0.32 mmol mL⁻¹, and were not significantly different among the longifolene treatments ($P > 0.05$). Only the highest concentration, 0.64 mmol mL⁻¹ treatment (82.41% $\pm$ 6.51%) resulted in a statistically significant

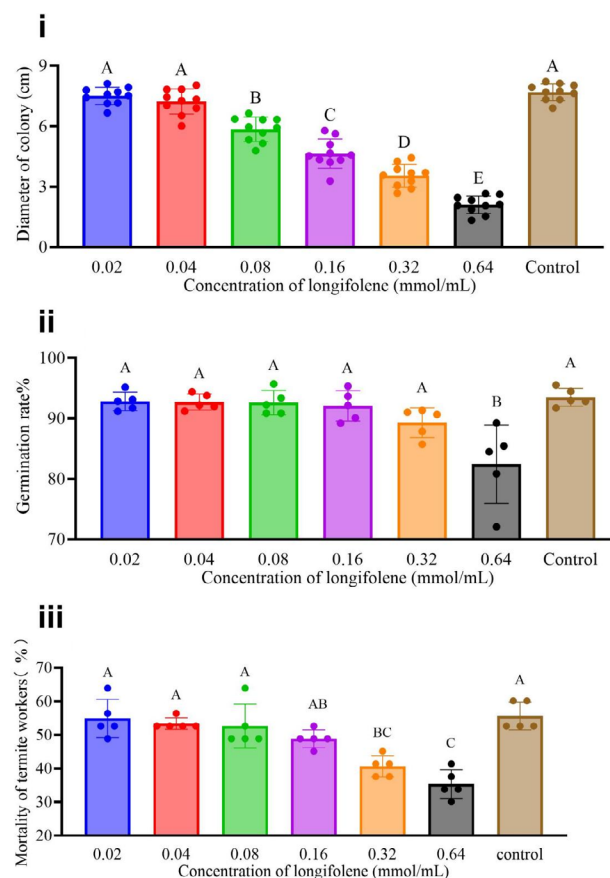


Figure 1. Effects of longifolene on the growth (i), germination (ii), and virulence-related characteristics (iii) of *Beauveria bassiana*. Statistical significance was evaluated using one-way analysis of variance (ANOVA), followed by Tukey's honest significant difference (HSD) *post hoc* test to identify significant differences between treatments ($P < 0.05$).

reduction in germination rate compared to the control (93.5% $\pm$ 1.49%) ($P < 0.0001$).

For virulence, the control group exhibited the highest corrected mortality (55.638% $\pm$ 4.12%), which was not significantly different from mortality rates observed at concentrations ≤ 0.16 mmol mL⁻¹ ($P > 0.05$, Figure 1(iii)). A moderate but significant reduction in virulence was observed at 0.32 mmol mL⁻¹ (40.598% $\pm$ 3.15%) ($P < 0.0001$), and the most pronounced reduction occurred at 0.64 mmol mL⁻¹ (35.336% $\pm$ 4.28%) ($P < 0.0001$). One-way ANOVA confirmed the significance of this trend ($F_{(6,28)} = 16.51$, $P < 0.0001$).

3.2 Dose-dependent repellency of *B. bassiana* dry conidia against *O. formosanus* workers

The repellency of *B. bassiana* dry conidia against *O. formosanus* workers exhibited a distinct dose-dependent pattern (Fig. 2). Behavioral assays revealed that while the control and 1×10^6 conidia mL⁻¹ treatments did not significantly influence workers' distribution ($P > 0.05$), concentrations ranging from 1×10^7 to 1×10^{10} conidia mL⁻¹ induced a robust avoidance response, with significantly more workers selecting the fresh-air arm (control) over the conidia-treated arm ($P < 0.001$; Fig. 2(ii)). Quantitative analysis further confirmed that repellent rates increased progressively with conidia concentration, rising from 6.88% $\pm$ 9.73% at 1×10^6 conidia mL⁻¹ to 59.37% $\pm$ 6.96% at 1×10^{10} conidia mL⁻¹ (Fig. 2(ii)). Two-way ANOVA and Tukey's HSD tests

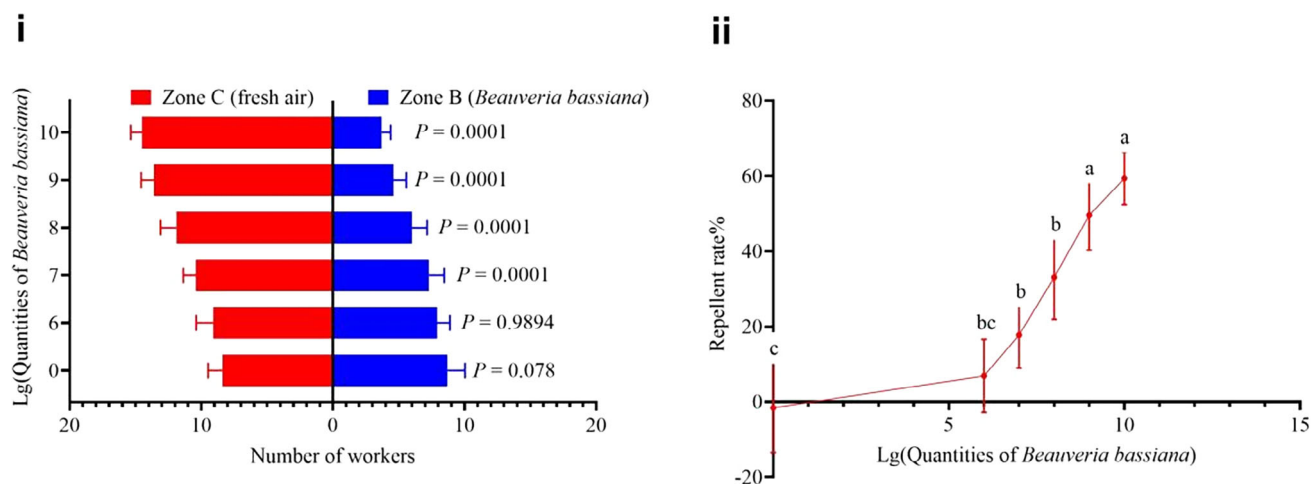


Figure 2. Repellent effects of *Beauveria bassiana* conidia on *Odontotermes formosanus* workers. (i) Numbers of workers choosing the treatment arm and control arm in the Y-tube olfactometer. (ii) Repellency rates of dry conidia at different concentrations. Different letters above bars indicate significant differences among treatments ($P < 0.05$).

demonstrated significant differences across most treatment levels, where moderate-to-high doses (≥ 0.04 g) yielded significantly higher repellency compared to both the control and low-concentration groups ($P < 0.05$). Collectively, these results indicate that *B. bassiana* dry conidia exerted a potent, dose-dependent repellency on *O. formosanus* workers (Fig. 2).

3.3 Attractant activity of longifolene on *O. formosanus* workers

Longifolene exhibited a significantly concentration-dependent attractant effect on *O. formosanus* workers (Fig. 3). While the solvent control (TEC) induced no behavioral preference ($P = 0.9999$), all longifolene treatments (0.05 – 0.8 mmol mL⁻¹) significantly increased termite recruitment to the treated arm compared to the control arm (fresh air) ($P = 0.0001$; Fig. 3(i)). The attractant rate followed a non-monotonic trend, rising from $24.68\% \pm 6.19\%$ at 0.05 mmol mL⁻¹ to a peak of $61.35\% \pm 13.44\%$ at 0.4 mmol mL⁻¹, before declining at 0.8 mmol mL⁻¹ (Fig. 3(ii)). Two-way ANOVA and Tukey's HSD tests confirmed that the attractant rates at 0.2 and

0.4 mmol mL⁻¹ were significantly higher than those of all other concentration groups ($P < 0.05$). These results demonstrate that longifolene effectively modulates the directional behavior of *O. formosanus* workers, with optimal attractant activity observed between 0.2 and 0.4 mmol mL⁻¹ (Fig. 3).

3.4 Attractant and repellency of longifolene combined with different concentrations of *B. bassiana* on *O. formosanus* workers

The combination of longifolene with *B. bassiana* at 1×10^7 conidia mL⁻¹ elicited a high concentration-dependent attractant effect on *O. formosanus* workers (Fig. 4(i)). The attractant rates increased from $0.99\% \pm 11.15\%$ in the control (without longifolene) to a peak of $71.06\% \pm 8.50\%$ at 0.4 mmol mL⁻¹ of longifolene, before slightly declining to $64.6\% \pm 7.65\%$ at 0.8 mmol mL⁻¹ of longifolene (Fig. 4(i)). Statistical analyses confirmed significant differences ($P < 0.05$), with the 0.4 and 0.8 mmol mL⁻¹ groups showing higher attractant rates than lower concentrations of longifolene ($P < 0.05$). In contrast, at 1×10^9 conidia mL⁻¹, a repellency

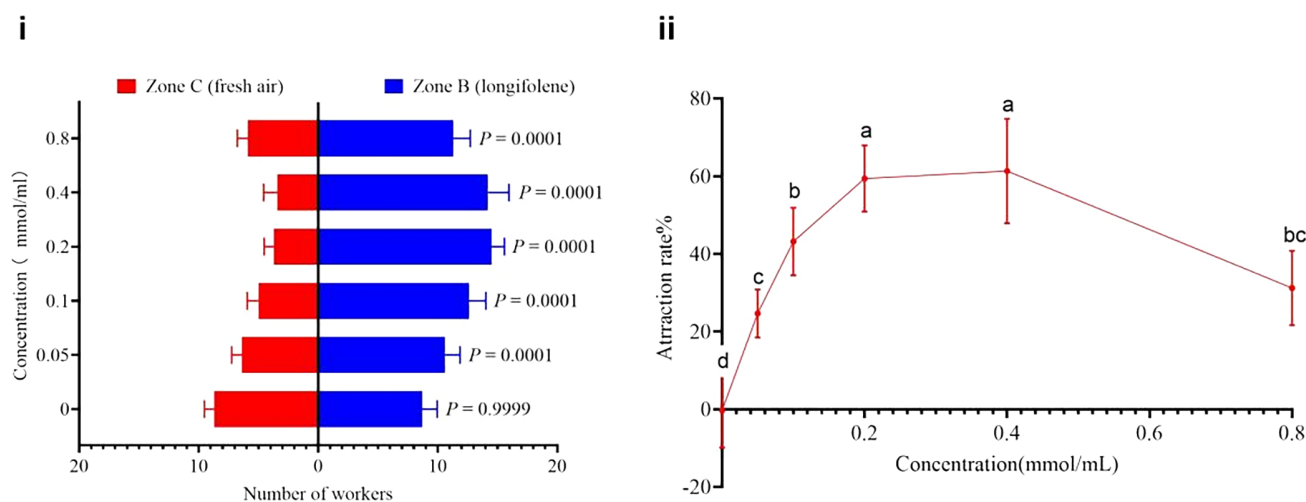


Figure 3. Attractant effects of longifolene on *Odontotermes formosanus* workers. (i) Numbers of workers choosing the longifolene arm and control arm in the Y-tube olfactometer. (ii) Attraction rates of longifolene at different concentrations. Different letters above bars indicate significant differences among treatments.

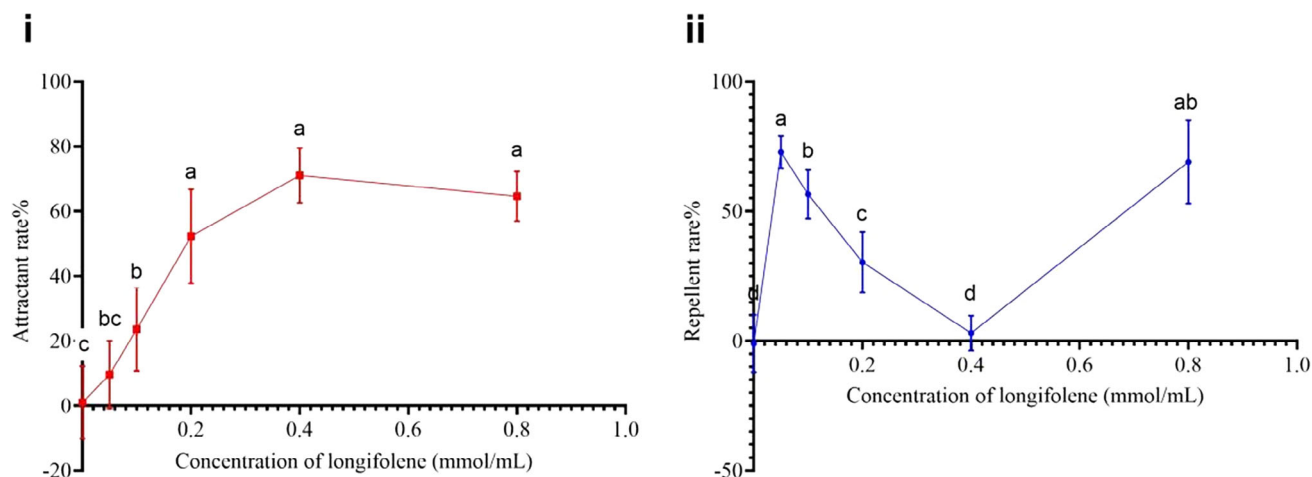


Figure 4. The attraction or repellent rates of the blend toxic bait. (i) The attraction efficacy of longifolene mixed with the lower concentration (1×10^7 conidia mL⁻¹) of *Beauveria bassiana* bait on termite workers. (ii) The repellent efficacy of longifolene mixed with the higher concentration (1×10^9 conidia mL⁻¹) of *B. bassiana* bait on termite workers.

predominated, peaking at $72.77\% \pm 6.23\%$ at 0.05 mmol mL⁻¹ longifolene and rebounding to $69.05\% \pm 16.05\%$ at 0.8 mmol mL⁻¹, while declining at intermediate concentrations (Fig. 4(ii)). Statistical analyses revealed significant differences ($P < 0.05$), with the 0.05 and 0.8 mmol mL⁻¹ groups exhibiting stronger repellency than other concentrations ($P < 0.05$).

3.5 Differential feeding attraction efficacy of longifolene-*B. bassiana* combined toxic baits at different conidia concentrations on *O. formosanus* workers in reservoir habitats

Field evaluations across the three reservoirs (R1–R3) sites revealed divergent performance between the two *B. bassiana*-based toxic baits (Toxic Bait 1 and Toxic Bait 2), which were selected based on the maximal laboratory attractancy (Fig. 5). In all the three reservoirs sites, Toxic Bait 1 (a combined formulation with 1×10^7 conidia mL⁻¹ *B. bassiana*, exhibiting the highest laboratory attractant rate) consistently demonstrated a significantly higher feeding attraction percentage compared to the control bait, with 82.81%

$\pm 11.10\%$ in R1, $73.11\% \pm 7.91\%$ in R2, and $71.35\% \pm 7.86\%$ in R3, while the corresponding control groups showed $17.19\% \pm 11.10\%$, $26.89\% \pm 7.91\%$, and $28.65\% \pm 7.86\%$, respectively (Fig. 5(i)); as also confirmed by independent-samples *t*-tests (with all the *P* values < 0.0001). Conversely, Toxic Bait 2 (a combined formulation with 1×10^9 conidia mL⁻¹ *B. bassiana*, also with the highest laboratory attractant rate) exhibited a lower feeding attraction percentage than the control bait across all the three reservoirs sites, with no significant differences observed, $44.39\% \pm 11.97\%$ in R1, $47.71\% \pm 4.51\%$ in R2, and $43.77\% \pm 6.29\%$ in R3, compared to $55.61\% \pm 11.97\%$, $52.29\% \pm 4.51\%$, and $56.23\% \pm 6.29\%$ in the control groups, respectively (Fig. 5(ii)). The statistical analysis revealed no significant differences ($P > 0.05$) between Toxic Bait 2 and the control in R1 ($P = 0.1195$), R2 ($P = 0.7700$), and R3 ($P = 0.0728$) (Fig. 5(ii)).

4 DISCUSSION

This study explored the potential of longifolene as a behavioral enhancer in *B. bassiana*-based bait systems targeting *O. formosanus*.

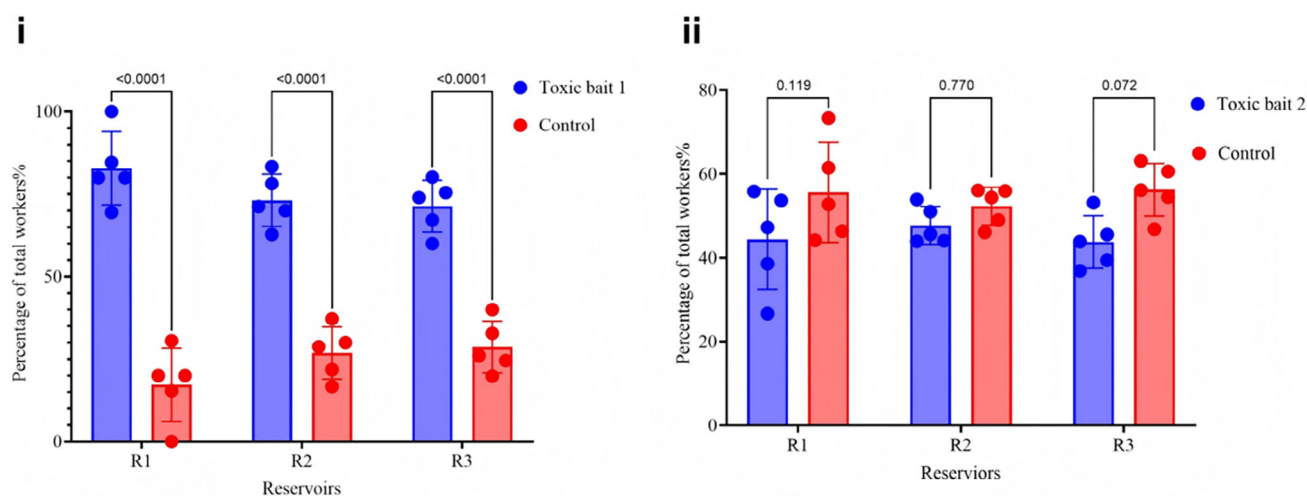


Figure 5. Field trial results for longifolene-supplemented toxic baits. (i) Toxic bait mixed with 1×10^7 conidia mL⁻¹ *Beauveria bassiana* conidial suspension (Toxic Bait 1). (ii) Toxic bait mixed with 1×10^9 conidia mL⁻¹ *B. bassiana* conidial suspension (Toxic Bait 2). One-way analysis of variance (ANOVA) was used to compare the difference, with Tukey's honest significant difference (HSD) test ($P < 0.05$). Numbers above the bars are *P* values.

Our results showed a significant concentration-dependent inhibitory effect of longifolene on the fungus, with substantial growth reduction observed at concentrations ≥ 0.04 mmol mL⁻¹. This finding is consistent with the previously reported antimicrobial properties of longifolene. For example, longifolene has been identified as a component with moderate antifungal activity in comparative quantitative structure–activity relationship analyses and has been isolated from *Juniperus communis* roots, where it induced antimycobacterial effects.²⁰ However, these previous studies did not systematically quantify the dose–response relationship of longifolene against a specific entomopathogenic fungus viability and efficacy. Our research fills this gap by clearly defining an inhibitory threshold of longifolene towards *B. bassiana*. Notably, at lower concentrations (≤ 0.01 mmol mL⁻¹), longifolene did not significantly hinder fungal growth, suggesting a range where its attractiveness function could be incorporated into fungal formulations without compromising pathogen viability and growth.

Our findings on the repellent response of *O. formosanus* workers to *B. bassiana* dry conidia expand upon earlier reports of fungal avoidance in termites by providing a detailed, dose-dependent characterization of this behavior.⁶ While previous research established a positive correlation between fungal virulence and repellency in *Macrotermes michaelseni*,⁸ our study systematically quantified repellency across a series of conidia concentration (10^7 – 10^{10} conidia mL⁻¹) that limited termite exposure to fungal volatile cues. This approach not only confirmed that olfactory detection mediates avoidance in termites but also revealed a continuous increase in repellency rate with conidia dose/concentration. Moreover, by employing a standardized plant-based carrier and preventing direct contact with conidia, our bioassay effectively established the olfactory component of avoidance, thereby offering a refined method for assessing semiochemical-based repellency in subterranean termites. These methodological and analytical advancements underscore the importance of dose–response profiling in understanding termite–fungus interactions and support the development of behavior-guided termite management strategies that utilize both repellent and attractant semiochemicals.

The study also demonstrate that longifolene acts as a significant attractant to *O. formosanus* workers within a concentration range of 0.2–0.4 mmol mL⁻¹. This finding aligns with its role as a broad-spectrum semiochemical in other biological systems. For instance, longifolene functions as an attractant in engineered fungal biocontrol strategies targeting mosquitoes,²¹ serves as a key component in host-location blends for pinewood nematodes,²² and is recognized as a ligand for odorant-binding proteins in parasitoid wasps.²³ These collective reports highlight the potential of longifolene as a reliable foraging or host-seeking cue across diverse taxa including Termitidae. In contrast, a previous study reported repellent and anti-termite activity of longifolene and its autooxidation products against *Reticulitermes speratus*.¹⁷ The discrepancy between our results and previous reports of longifolene repellency against *R. speratus* may be attributable to differences in termite ecology, experimental design, or chemical purity. Different termite species may vary in their olfactory sensitivity and ecological preferences, and behavioral responses can also be influenced by test system configuration, volatile release dynamics, and the source or purity of the compound used. Therefore, the attractant effect observed here should be interpreted within the context of *O. formosanus* biology and the specific bioassay conditions employed in this study. Our results clarify the attractant function of pure longifolene at specified concentrations for *O. formosanus*,

supporting its potential integration with low-concentration of *B. bassiana* bait to develop an effective ‘attract-and-infect’ strategy for subterranean termite biological control.

To assess the applicability of the longifolene-mediated behavioral effects under natural conditions, we conducted a field trial at reservoir sites using longifolene-supplemented toxic baits (Toxic Bait 1 and Toxic Bait 2). Over the 20-day period, both bait formulations attracted significantly more termite workers than the control baits, with feeding attraction exceeding 71%, indicating that longifolene can promote bait acceptance in the field and potentially facilitate an ‘attract-and-infect’ process. In the present field study, we quantified feeding attraction (bait acceptance), but colony-level suppression and individual mortality were not measured or confirmed. Feeding attraction indicates whether termites accept and recruit to bait stations, yet it does not directly demonstrate infection or lethal efficacy. Future studies should include termite recovery from bait stations, assessment of *B. bassiana* infection, and monitoring over longer periods to evaluate colony-level effects.

Although our results clearly show that longifolene enhances termite attraction and reverses the repellent/avoidance effect associated with *B. bassiana* conidia, the causal mechanism was not directly tested in this study. Based on existing literature on semiochemical interactions, longifolene may counteract fungal repellency either by (i) acting as an independent attractant that increases foraging motivation and thereby outweighs avoidance, and/or (ii) modulating the olfactory cues by masking or competing with repellent volatiles emitted from the fungus in the bait headspace. We therefore propose that the combined formulation may facilitate an ‘attract-and-infect’ process, but future mechanistic work is needed to confirm whether masking/olfactory competition or motivational override is the dominant driver.

Our investigation into the combined use of *B. bassiana* and longifolene in a bait formulation represents a novel approach to addressing a key limitation in fungal-based biopesticide application for termite control – namely, the repellency or avoidance of high-conidium doses. Previous studies have shown that semiochemicals can be used to improve the performance of entomopathogenic fungi by increasing host attraction, reducing avoidance, or facilitating pathogen contact.^{24,25} In this context, our results extend the semiochemical-assisted biocontrol concept by showing that longifolene not only attracts *O. formosanus* workers but also counteracts the repellency associated with *B. bassiana* conidia. This suggests that combining a fungal pathogen with an attractive volatile cue may be a more effective strategy than relying on fungal inoculum alone, particularly for cryptic pests that exhibit strong avoidance behavior. Similar naturally derived attractant approach was also demonstrated by a previous study²⁶ while co-formulating carbon dioxide (CO₂)-emitting capsules with *Metarhizium brunneum* Cb15-III to mitigate subterranean termite avoidance. With this co-formulation technique, the authors reported termite workers' infection by the fungus growing from the capsules as well as its horizontal transmission through an autodissemination approach.²⁶ Our results clearly showed that longifolene, when combined with a low concentration of *B. bassiana* (1×10^7 conidia mL⁻¹), not only counteracted the inherent repellency of the fungus but also produced an attractant rate exceeding 70% in laboratory assays. This ‘attract-and-infect’ effect was further validated under field conditions, where the longifolene-supplemented bait achieved a feeding attraction percentage above 70% across multiple reservoir sites, significantly outperforming the control. These findings suggest that

integrating a behavior-modifying semiochemical like longifolene could effectively mitigate the dose-dependent repellency or avoidance that often hinders the field performance of entomopathogenic fungi. By enhancing bait palatability and ensuring sustained termite exposure to infective conidia, this combined formulation addresses a critical gap in current baiting technology. Future bait development efforts could therefore benefit from incorporating such dual-function blends, which synergistically exploit termite olfactory ecology and fungal pathogenicity to achieve more sustainable and effective termite management.

5 CONCLUSION

Our findings indicate that longifolene can mitigate the olfactory repellency of *B. bassiana* conidia in baits and thereby facilitate an attract-and-infect strategy against *O. formosanus*. Laboratory growth-related assays suggested that longifolene did not obviously inhibit fungal performance under the tested conditions; however, fungal pathogenicity was not directly evaluated in this study and should be confirmed in future bioassays.

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DATA AVAILABILITY STATEMENT

The data that support the findings of this study are openly available in Longifolene Enhances Beauveria bassianaBait by Counteracting at <http://10.6084/m9.figshare.32214498>.

CONFLICT OF INTEREST

The authors declare no competing interests.

SUPPORTING INFORMATION

Supporting information may be found in the online version of this article.

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